In [9]:
```
import matplotlib.pyplot as plt
import numpy as np
```

In [10]:
```
# Cost Function

def cost_function(x) :
    return (x+3)**2
```

In [11]:
```
# Gradient of the Function
# (Derivative)

def gradient(x) :
    return 2*(x+3)
```

In [12]:
```
x_values = []
y_values = []
num_iterations = 100
```

In [13]:
```
x_initial = 2
x = x_initial
learning_rate = 0.1
```

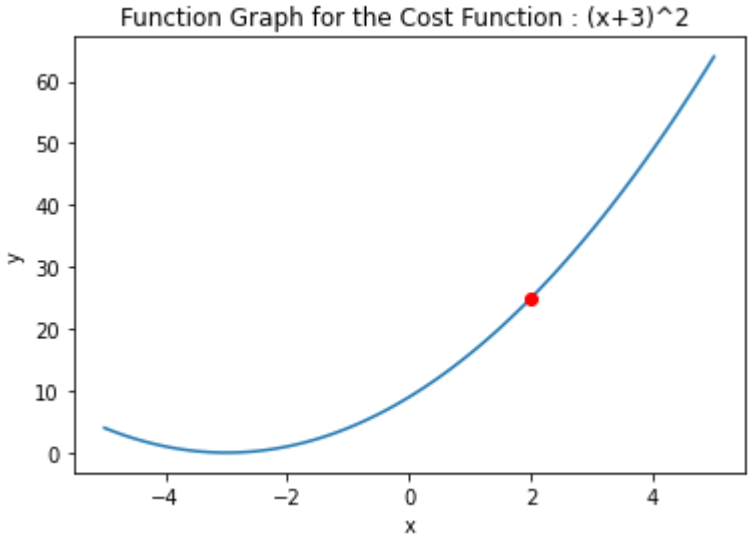
In [14]:
```
for i in range(num_iterations) :
    x_values.append(x)
    y_values.append(cost_function(x))
    x = x - learning_rate * gradient(x)
    print(f'Iteration {i+1} : x Value : {x}, Cost : {cost_function(x)}')
print(f'Optimal x : {x}')
```

```
Iteration 1 : x Value : 1.0, Cost : 16.0
Iteration 2 : x Value : 0.19999999999999996, Cost : 10.240000000000002
Iteration 3 : x Value : -0.44000000000000017, Cost : 6.553599999999998
Iteration 4 : x Value : -0.9520000000000001, Cost : 4.194304
Iteration 5 : x Value : -1.3616000000000001, Cost : 2.6843545599999996
Iteration 6 : x Value : -1.6892800000000001, Cost : 1.7179869183999996
Iteration 7 : x Value : -1.951424, Cost : 1.099511627776
Iteration 8 : x Value : -2.1611392, Cost : 0.7036874417766399
Iteration 9 : x Value : -2.32891136, Cost : 0.4503599627370493
Iteration 10 : x Value : -2.463129088, Cost : 0.28823037615171165
Iteration 11 : x Value : -2.5705032704, Cost : 0.1844674407370954
Iteration 12 : x Value : -2.6564026163200003, Cost : 0.11805916207174093
Iteration 13 : x Value : -2.725122093056, Cost : 0.07555786372591429
Iteration 14 : x Value : -2.7800976744448, Cost : 0.04835703278458515
Iteration 15 : x Value : -2.82407813955584, Cost : 0.030948500982134555
Iteration 16 : x Value : -2.8592625116446717, Cost : 0.019807040628566166
Iteration 17 : x Value : -2.8874100093157375, Cost : 0.012676506002282305
Iteration 18 : x Value : -2.90992800745259, Cost : 0.008112963841460692
Iteration 19 : x Value : -2.927942405962072, Cost : 0.005192296858534868
Iteration 20 : x Value : -2.9423539247696575, Cost : 0.0033230699894623056
Iteration 21 : x Value : -2.953883139815726, Cost : 0.002126764793255884
Iteration 22 : x Value : -2.9631065118525806, Cost : 0.0013611294676837786
Iteration 23 : x Value : -2.9704852094820646, Cost : 0.0008711228593176078
Iteration 24 : x Value : -2.9763881675856516, Cost : 0.0005575186299632732
Iteration 25 : x Value : -2.981110534068521, Cost : 0.0003568119231765016
Iteration 26 : x Value : -2.984888427254817, Cost : 0.0002283596308329556
Iteration 27 : x Value : -2.9879107418038537, Cost : 0.00014615016373308945
Iteration 28 : x Value : -2.990328593443083, Cost : 9.353610478917726e-05
Iteration 29 : x Value : -2.992262874754466 4, Cost : 5.986310706507345e-05
Iteration 30 : x Value : -2.993810299803573, Cost : 3.83123885216492e-05
Iteration 31 : x Value : -2.995048239842858, Cost : 2.451992865385725e-05
Iteration 32 : x Value : -2.9960385918742864, Cost : 1.5692754338469342e-05
Iteration 33 : x Value : -2.9968308734994293, Cost : 1.0043362776619253e-05
Iteration 34 : x Value : -2.9974646987995435, Cost : 6.427752177036323e-06
Iteration 35 : x Value : -2.997971759039635, Cost : 4.113761393302886e-06
Iteration 36 : x Value : -2.998377407231708, Cost : 2.6328072917135587e-06
Iteration 37 : x Value : -2.998701925785366, Cost : 1.6849966666971388e-06
Iteration 38 : x Value : -2.998961540628293, Cost : 1.0783978666865378e-06
Iteration 39 : x Value : -2.9991692325026342, Cost : 6.901746346793842e-07
Iteration 40 : x Value : -2.9993353860021075, Cost : 4.417117661946878e-07
Iteration 41 : x Value : -2.999468308801686, Cost : 2.826955303647891e-07
Iteration 42 : x Value : -2.9995746470413485, Cost : 1.8092513943361614e-07
Iteration 43 : x Value : -2.9996597176330786, Cost : 1.1579208923763523e-07
Iteration 44 : x Value : -2.999727774106463, Cost : 7.410693711203819e-08
Iteration 45 : x Value : -2.99978221928517, Cost : 4.7428439751781807e-08
Iteration 46 : x Value : -2.9998257754281363, Cost : 3.035420144107846e-08
```

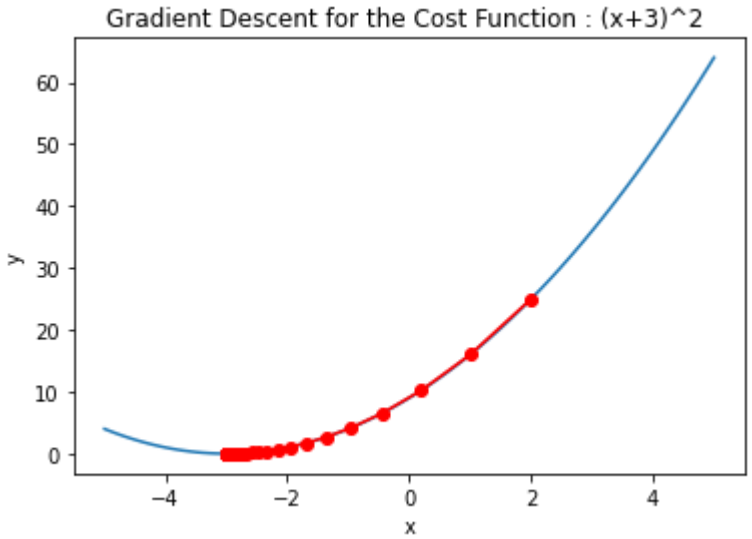
```
Iteration 47 : x Value : -2.999860620342509, Cost : 1.9426688922339734e-08
Iteration 48 : x Value : -2.999888496274007, Cost : 1.243308091029743e-08
Iteration 49 : x Value : -2.9999107970192056, Cost : 7.9571717826062e-09
Iteration 50 : x Value : -2.9999286376153647, Cost : 5.092589940842615e-09
Iteration 51 : x Value : -2.9999429100922916, Cost : 3.259257562149415e-09
Iteration 52 : x Value : -2.999954328073833, Cost : 2.0859248397837384e-09
Iteration 53 : x Value : -2.9999634624590668, Cost : 1.3349918974486118e-09
Iteration 54 : x Value : -2.9999707699672533, Cost : 8.543948143723039e-10
Iteration 55 : x Value : -2.999976615973803, Cost : 5.468126811899669e-10
Iteration 56 : x Value : -2.9999812927790424, Cost : 3.499601159582557e-10
Iteration 57 : x Value : -2.9999850342232337, Cost : 2.2397447421860056e-10
Iteration 58 : x Value : -2.999988027378587, Cost : 1.433436634977776e-10
Iteration 59 : x Value : -2.9999904219028695, Cost : 9.173994464198049e-11
Iteration 60 : x Value : -2.9999923375222957, Cost : 5.871356456950638e-11
Iteration 61 : x Value : -2.9999938700178364, Cost : 3.757668132666189e-11
Iteration 62 : x Value : -2.999995096014269, Cost : 2.4049076048192486e-11
Iteration 63 : x Value : -2.9999960768114153, Cost : 1.5391408670843192e-11
Iteration 64 : x Value : -2.9999968614491324, Cost : 9.850501548782124e-12
Iteration 65 : x Value : -2.999997489159306, Cost : 6.3043209907745444e-12
Iteration 66 : x Value : -2.9999979913274446, Cost : 4.034765434809332e-12
Iteration 67 : x Value : -2.9999983930619556, Cost : 2.5822498785634223e-12
Iteration 68 : x Value : -2.9999987144495646, Cost : 1.6526399220522305e-12
Iteration 69 : x Value : -2.9999989715596516, Cost : 1.0576895502961154e-12
Iteration 70 : x Value : -2.9999991772477212, Cost : 6.769213121895138e-13
Iteration 71 : x Value : -2.999999341798177, Cost : 4.3322963956744853e-13
Iteration 72 : x Value : -2.9999994734385416, Cost : 2.7726696951023927e-13
Iteration 73 : x Value : -2.9999995787508333, Cost : 1.7745086041172427e-13
Iteration 74 : x Value : -2.9999996630006667, Cost : 1.1356855066350352e-13
Iteration 75 : x Value : -2.9999997304005332, Cost : 7.268387247253274e-14
Iteration 76 : x Value : -2.9999997843204267, Cost : 4.651767834410857e-14
Iteration 77 : x Value : -2.9999998274563415, Cost : 2.977131407892966e-14
Iteration 78 : x Value : -2.9999998619650734, Cost : 1.9053640961475125e-14
Iteration 79 : x Value : -2.9999998895720585, Cost : 1.2194330254575965e-14
Iteration 80 : x Value : -2.999999911657647, Cost : 7.804371331543109e-15
Iteration 81 : x Value : -2.9999999293261177, Cost : 4.994797639633387e-15
Iteration 82 : x Value : -2.999999943460894, Cost : 3.1966704893653676e-15
Iteration 83 : x Value : -2.9999999547687155, Cost : 2.045869097124455e-15
Iteration 84 : x Value : -2.9999999638149726, Cost : 1.309356209304147e-15
Iteration 85 : x Value : -2.9999999710519782, Cost : 8.379879636702507e-16
Iteration 86 : x Value : -2.9999999768415826, Cost : 5.363122967489605e-16
Iteration 87 : x Value : -2.999999981473266, Cost : 3.432398699193347e-16
Iteration 88 : x Value : -2.9999999851786128, Cost : 2.1967351938118145e-16
Iteration 89 : x Value : -2.99999998814289, Cost : 1.4059105661644777e-16
Iteration 90 : x Value : -2.999999990514312, Cost : 8.997827286453327e-17
Iteration 91 : x Value : -2.9999999924114498, Cost : 5.758609463330129e-17
Iteration 92 : x Value : -2.9999999939291597, Cost : 3.685510164371068e-17
Iteration 93 : x Value : -2.9999999951433276, Cost : 2.358726677741145e-17
Iteration 94 : x Value : -2.999999996114662, Cost : 1.5095850047368678e-17
Iteration 95 : x Value : -2.99999999689173, Cost : 9.661342926036557e-18
Iteration 96 : x Value : -2.9999999975133838, Cost : 6.18326035608692e-18
Iteration 97 : x Value : -2.999999998010707, Cost : 3.957287334634505e-18
Iteration 98 : x Value : -2.9999999984085655, Cost : 2.532663894166083e-18
Iteration 99 : x Value : -2.999999998726852, Cost : 1.6209053445792253e-18
```

```
Iteration 100 : x Value : -2.999999998981482, Cost : 1.0373792396055266e-18
Optimal x : -2.999999998981482
```

In [24]:
```
x_coordinates = np.linspace(-5,5,100)
plt.plot(x_coordinates,cost_function(x_coordinates))
plt.xlabel("x")
plt.ylabel("y")
plt.title("Function Graph for the Cost Function : (x+3)^2 ")
plt.plot(2,cost_function(2),'ro')
plt.show()
# Red Dot indicates the Starting Value of x = 2
```



In [25]:
```
plt.plot(x_coordinates,cost_function(x_coordinates))
plt.plot(x_values,y_values,'ro-')
plt.xlabel("x")
plt.ylabel("y")
plt.title("Gradient Descent for the Cost Function : (x+3)^2 ")
plt.show()
```



In [ ]:
```
 # Hence we have Found Gradient Descent of the Algorithm and Found the Local Minimum
```